

Expectations and the Rate of Inflation

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Introduction

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 3. Is it **long-run** or **short-run** expectations that matter the most?
 4. How does **past inflation** impact on current inflation?

- ▶ The goal is to isolate the current impact of a change in expectations, holding everything else fixed (departure from rational expectations).

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Passthrough potentially very low.

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It is expectations in the near future that matters the most.

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The long-run Phillips curve is vertical.
 4. **General State Dependent -pricing:**
Standard menu-cost models produce extreme results. Consider two variations.

Calvo vs Taylor

Calvo Pricing

Calvo vs Taylor

Calvo Pricing

- Constant hazard rate of changing prices: $1 - \lambda$. Log-lin optimal reset price:

$$p_t^* = \mu + (1 - \beta\lambda)\mathbb{E}_{t-1} \sum_{s=0}^{\infty} (\beta\lambda)^s (P_{t+s} + mc_{t+s}) \quad (1)$$

Firms set their price so that they are at their ideal value $(\mu + P_{t+s} + mc_{t+s})$ on average.

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- To focus on the contribution from inflation expectations:

$$p_t^* - P_{t-1} = (1 - \beta\lambda)\mathbb{E}_{t-1} \sum_{s=0}^{\infty} (\beta\lambda)^s (P_{t+s} - P_{t-1}) + a_t \quad (2)$$

where $a_t \equiv \mu + \mathbb{E}_{t-1} \sum_{s=0}^{\infty} (\beta\lambda)^s mc_{t+s}$ collects all the non inflation expectation items.

Calvo vs Taylor

Calvo Pricing

- ▶ Consider expectation of future expectation that are constant at some π^e across all horizons: $\mathbb{E}_{t-1}P_{t+s} - P_{t-1} = \pi^e(1 + s)$

Calvo vs Taylor

Calvo Pricing

- ▶ Consider expectation of future expectation that are constant at some π^e across all horizons: $\mathbb{E}_{t-1}P_{t+s} - P_{t-1} = \pi^e(1 + s)$
- ▶ Substituting this constant expectation into the price setting condition:

$$p_t^* - P_{t-1} = (1 - \beta\lambda) \sum_{s=0}^{\infty} (\beta\lambda)^s (1 + s) \pi^e + a_t = \frac{1}{1 - \beta\lambda} \pi^e + a_t \quad (3)$$

As long as prices are sticky ($\lambda > 0$) the coefficient on π^e is greater than one: **firms overshoot their price**

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As long as prices are sticky ($\lambda > 0$) the coefficient on π^e is greater than one: **firms overshoot their price**

As $\lambda \rightarrow 0$ or as $\beta \rightarrow 0$, the coefficient becomes one: firms do not overshoot their price, and simply set it at the static optimum $p_t^* = P_t^e = P_{t-1} + \pi^e$.

As $\beta\lambda \rightarrow 1$ the overshooting becomes infinitely large.

Calvo vs Taylor

Calvo Pricing

- Inflation is the weighted average of inflation across firms that cannot change their prices (0 with weight λ) and those that do reset prices ($p_t^* - P_{t-1}$ with weight $1 - \lambda$):

$$\pi_t \equiv P_t - P_{t-1} = (1 - \lambda) (p_t^* - P_{t-1}) \quad (4)$$

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- Combined with the prev. result (eq (3)):

$$\pi_t = \phi \pi^e + (1 - \lambda) a_t \quad (5)$$

where

$$\phi = \frac{1 - \lambda}{1 - \beta\lambda} \quad (6)$$

Calvo vs Taylor

Calvo Pricing

► **Proposition 1** : In the Calvo model passthrough ϕ satisfies:

$$\phi \in (0, 1)$$

and any value can be attained in this interval for some parameters.

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- In the limit of no discounting $\beta \rightarrow 1$ or flexible prices $\lambda \rightarrow 0$: then $\phi \rightarrow 1$
- In the limit of rigid prices $\lambda \rightarrow 1$ then $\phi \rightarrow 0$.

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- In the limit of rigid prices $\lambda \rightarrow 1$ then $\phi \rightarrow 0$.
- The limit with no discounting giving $\phi = 1$ is a good approximation in the Calvo model.

Calvo vs Taylor

Taylor Pricing

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- ▶ A fraction $1/N$ of firms changes prices each period and reset prices every N periods.

Calvo vs Taylor

Taylor Pricing

- ▶ A fraction $1/N$ of firms changes prices each period and reset prices every N periods.
- ▶ Reset price as a weighted average of the expected prices

$$p_t^* = \frac{\sum_{s=0}^{N-1} \beta^s P_{t+s}^e}{\sum_{s=0}^{N-1} \beta^s} + a_t \quad (7)$$

where $a_t \equiv \mu + \mathbb{E}_{t-1} \sum_{s=0}^{N-1} \beta^s mc_{t+s} / \sum_{s=0}^{N-1} \beta^s$.

Calvo vs Taylor

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- ▶ Again, with constant expectations of future inflation across all horizons:

$$p_t^* - P_{t-1} = \frac{\sum_{s=0}^{N-1} \beta^s (s+1)}{\sum_{s=0}^{N-1} \beta^s} \pi^e + a_t \quad (8)$$

Calvo vs Taylor

Taylor Pricing

- Inflation is the weighted average of inflation across firms that cannot change their prices (0 with weight $(N - 1)/N$) and those that do reset prices ($p_t^* - P_{t-1}$ with weight $1/N$):

$$\pi_t = \frac{1}{N} (p_t^* - P_{t-1}) \quad (9)$$

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Taylor Pricing

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- Combined with the prev. result (eq (8)):

$$\pi_t = \phi \pi^e + \frac{1}{N} a_t \quad (10)$$

with

$$\phi = \frac{1}{N} \frac{\sum_{s=0}^{N-1} \beta^s (s+1)}{\sum_{s=0}^{N-1} \beta^s} \quad (11)$$

Calvo vs Taylor

Taylor Pricing

- **Proposition 2** : In the Taylor model passthrough ϕ satisfies:

$$\phi \in (0, 1/2)$$

Calvo vs Taylor

Taylor Pricing

- ▶ **Proposition 2** : In the Taylor model passthrough ϕ satisfies:

$$\phi \in (0, 1/2)$$

- ▶ In the limit of no discounting or flexible prices $\phi \rightarrow 1/2$.
- ▶ In the limit of rigid prices $\phi \rightarrow 0$.
- ▶ The no discounting case is a good approximation for reasonable parameters.

General TD Model

General Time Dependent Pricing

General TD Model

General Time Dependent Pricing

- ▶ Hazard function (h_s): probability of re-setting price $s + 1$ periods since the previous reset
- ▶ Survival probability (S_s) for each age s : $S_{s+1} = S_s (1 - h_s)$, with $S_0 = 1$.
- ▶ $F_s = 1 - S_{s+1}$ (with $F_{-1} = 0$) represents the probability a spell will be s or less. The associated density $f_s = S_s h_s$ gives the distributions of completed spells.
- ▶ Define the distribution of ongoing spells, ω_s :

$$\omega_s = \frac{S_s}{\sum_{s=0}^{\infty} S_s} \quad (12)$$

captures the relative importance of age s relative to all other ages .

General TD Model

General Time Dependent Pricing

- ▶ **Firm's Problem.**

faces a path of θ_t shocks to its profit function

- ▶ A firm resetting its price in period t then solves:

$$\max_{p_t^*} \mathbb{E}_{t-1} \sum_{s=0}^{\infty} \beta^s \omega_s \Pi(p_t^* - P_{t+s}, \theta_{t+s}) \quad (13)$$

- ▶ **Proposition 3:** To a first-order approximation, around a zero inflation steady state, the reset price satisfies:

$$p_t^* = \frac{\mathbb{E}_{t-1} \sum_{s=0}^{\infty} \beta^s \omega_s P_t^e}{\mathbb{E}_{t-1} \sum_{s=0}^{\infty} \beta^s \omega_s} + a_t(\theta_t)$$

- ▶ The analysis takes the sequence for $\{\theta_t\}$ and $\{a_t\}$ as given and is focused on the effects of expectations $\{P_t^e\}$.

General TD Model

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- ▶ **Goal:** calculate the impact of a sudden change in inflation expectations.

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General Time Dependent Pricing

- **Goal:** calculate the impact of a sudden change in inflation expectations.
- Approximate around a zero inflation rate steady state and assume no discounting limit $\beta \rightarrow 1$:

$$p_t^* = \sum_{s=0}^{\infty} \omega_s P_{t+s}^e = \sum_{s=0}^{\infty} \omega_s \pi^e (1 + s) + P_{t-1} \quad (14)$$

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- ▶ **Goal:** calculate the impact of a sudden change in inflation expectations.
- ▶ Approximate around a zero inflation rate steady state and assume no discounting limit $\beta \rightarrow 1$:

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- ▶ Assume inflation has been zero in the past and that all firms have the initial price P_{t-1}
- ▶ All firms set price to p_t^* at $t = 0$ when the shock to expectations occurs, so

$$\pi_t = \bar{h} (p_t^* - P_{t-1})$$

General TD Model

General Time Dependent Pricing

- **Proposition 4:** Up to a first-order approximation around a zero inflation steady state with zero past inflation

$$\pi_t = \phi \pi^e + a_t \quad (15)$$

where

$$\phi = \bar{h} \sum_{s=0}^{\infty} \omega_s (1+s) = \frac{\sum_{s=0}^{\infty} \omega_s (1+s)}{\sum_{s=0}^{\infty} f_s (1+s)} = \frac{\hat{d}}{\bar{d}} \quad (16)$$

- The passthrough ϕ equals the ratio of the duration of ongoing spells $\hat{d}$ to that of completed spells $\bar{d}$.
1. The duration of ongoing spells controls the incentive firms have to overshoot their price, relative to their current ideal price, in the face of expected inflation.
 2. The duration of completed spells captures the frequency of price changes.

General TD Model

General Time Dependent Pricing

► **Proposition 5:** Let ϕ be given by Proposition 4 then for any h_s :

$$\phi \geq \frac{1}{2} \tag{17}$$

In particular, $\phi > 1$ arbitrarily large is possible for some h_s .

General TD Model

General Time Dependent Pricing

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In particular, $\phi > 1$ arbitrarily large is possible for some h_s .

- **Lower bound:** The duration of ongoing spells suffers from two “biases” that make it generally different from that of completed spells:
 1. for any given spell the age at which we sample an ongoing spell is by definition below that of the completed spell; a downward bias.
 2. unless spells are all of the same duration, we oversample relatively longer spells; an upward bias.
- **Upper bound:** average duration of ongoing spells can be very large relative to the completed ones by having most spells end as soon as they start, but have a small fraction end after a very long time.

General TD Model

General Phillips Curve

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General Phillips Curve

- ▶ Relax two assumptions:
 1. Expectations for inflation at different horizons are no longer constant but are given by an arbitrary sequence $\{\pi_{t+s}^e\}_{s=0}^{\infty}$
 2. Past realized inflation is no longer 0 but given by an arbitrary sequence $\{\pi_{t-s}\}_{s=0}^{\infty}$

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 1. Expectations for inflation at different horizons are no longer constant but are given by an arbitrary sequence $\{\pi_{t+s}^e\}_{s=0}^{\infty}$
 2. Past realized inflation is no longer 0 but given by an arbitrary sequence $\{\pi_{t-s}\}_{s=0}^{\infty}$
- ▶ **Proposition 7:** Suppose the hazard function is strictly positive $h_s > 0$ for all s then to a first-order approximation in the limit of no discounting we have:

$$\pi_t = \sum_{s=0}^{\infty} \phi_s \pi_{t+s}^e + \sum_{s=-1}^{-\infty} \phi_s \pi_{t+s} + a_t \quad (18)$$

with ϕ_s decreasing for $s > 0$ and

$$\sum_{s=-\infty}^{\infty} \phi_s = 1$$

and $\{a_t\}$ a sequence dependent on the expectation of future real variables.

General TD Model

General Phillips Curve

General TD Model

General Phillips Curve

- ▶ Reset price w/o imposing constant π^e :

$$p_t^* = \sum_{s=0}^{\infty} \omega_s P_{t+s}^e = \sum_{s=0}^{\infty} \omega_s \sum_{j=0}^s \pi_{t+j}^e + P_{t-1} = \sum_{s=0}^{\infty} (1 - \Omega_{s-1}) \pi_{t+s}^e + P_{t-1} \quad (19)$$

where the cumulative distribution is $\Omega_s = \sum_{n=0}^s \omega_n$ with the convention that $\Omega_{-1} = 0$.

- ▶ Using $\pi_t = \sum_{s=0}^{\infty} \omega_s h_s (p_t^* - p_{t-1-s}^*)$ [▶ Go](#) we arrive a

$$\begin{aligned} \pi_t &= \sum_{s=0}^{\infty} \omega_s h_s (p_t^* - P_{t-1}) + \sum_{s=0}^{\infty} \omega_s h_s (P_{t-1} - p_{t-1-s}^*) \\ &= \bar{h} \sum_{s=0}^{\infty} (1 - \Omega_{s-1}) \pi_{t+s}^e + \sum_{s=0}^{\infty} \omega_s (\bar{h} - h_s) p_{t-1-s}^* \end{aligned}$$

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General Phillips Curve

- ▶ The first set of terms provides the required coefficients $\phi_s = \bar{h} (1 - \Omega_{s-1})$ for expectations of future inflation $s \geq 0$.

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- ▶ ϕ_s are decreasing since the cumulative distribution Ω_{s-1} is increasing.
- ▶ When resetting their prices, **firms care relatively more about earlier inflation** because it affects the price level over more periods.

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- ▶ The second backward-looking set of terms depend on past reset prices:
$$\sum_{s=0}^{\infty} \omega_s (\bar{h} - h_s) p_{t-1-s}^*$$

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- ▶ When resetting their prices, **firms care relatively more about earlier inflation** because it affects the price level over more periods.
- ▶ The second backward-looking set of terms depend on past reset prices:
$$\sum_{s=0}^{\infty} \omega_s (\bar{h} - h_s) p_{t-1-s}^*$$
- ▶ If either $h_s = \bar{h}$ (as in Calvo) or $p_{t-1-s}^* = p^*$, then this backward-looking term is zero.
- ▶ If p_t^* has been rising in the past and the hazard rate h_s is increasing, then this new term contributes towards positive current inflation.

General SD Model

State Dependent Models

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- ▶ **Menu Cost Models.** Price adjustments is endogenous: sS bands characterizing regions of inaction where the price is left unchanged.

General SD Model

State Dependent Models

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- ▶ Cts time. Agg price $P_t = \int p_t^i di$, the price gap $x_t^i = p_t^i - P_t$.
- ▶ Firms profits are a function of their price gap $f(x)$. Discounting is constant, ρ .
- ▶ Fixed cost c each time they change their price: (s, S) rule keeping x in the interval $[s, S]$ with $s < 0 < S$ adjusting the price up to $x = S$ when $x = s$.

General SD Model

State Dependent Models

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- ▶ Firms profits are a function of their price gap $f(x)$. Discounting is constant, ρ .
- ▶ Fixed cost c each time they change their price: (s, S) rule keeping x in the interval $[s, S]$ with $s < 0 < S$ adjusting the price up to $x = S$ when $x = s$.
- ▶ **Goal:** characterizing the (s, S) rule and deriving its implications for the passthrough from expected inflation to realized inflation.

General SD Model

State Dependent Models

- ▶ Consider a steady state where expected inflation is $\pi^e > 0$.
- ▶ Then the firm problem at time t is:

$$V(x_t; \pi^e) \equiv \max_T \int_0^T e^{-\rho s} f(x_{t+s}) dt + e^{-\rho T} (V^*(\pi^e) - c) \quad (20)$$
$$x_{t+s} = x_t - \pi^e t$$

with x_t given and V^* being the stationary value of a firm that just optimally changed prices.

General SD Model

State Dependent Models

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$$x_{t+s} = x_t - \pi^e t$$

with x_t given and V^* being the stationary value of a firm that just optimally changed prices.

- ▶ From [Sheshinski and Weiss, 1977](#):
 1. The bands are then symmetric around zero: $\bar{x} = x^*, \underline{x} = -x^*$.
 2. For higher π^e , firms will expand their bands, so that $x^*(\pi^e) = -x^*(\pi^e)$ is increasing in π^e .
 3. For higher π^e , prices will be set for a shorter amount of time T .

General SD Model

State Dependent Models

- **Proposition 8** : In the Sheshinski-Weiss menu cost model, starting from a steady state with π_0 , and $\pi^e = \pi_0$:
1. an increase in inflation expectations at $t = 0$, $\pi^e > \pi_0$ lowers realized inflation on impact to zero $\pi = 0$.
 2. a decrease in inflation expectations $\pi^e < \pi_0$ induces at a mass of firms to change their price immediately: upward jump in the price level that is greater the larger the change in expectations.

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 2. a decrease in inflation expectations $\pi^e < \pi_0$ induces at a mass of firms to change their price immediately: upward jump in the price level that is greater the larger the change in expectations.
- ▶ For an increase in π^e the passthrough is initially negative infinity. Once inflation is at zero further increases in π^e have no effect, so the marginal passthrough becomes zero.
- ▶ For a decrease in π^e the immediate passthrough is also negative.

General SD Model

State Dependent Models

- ▶ The results above are extreme and probably not realistic: **Two Extensions:**

General SD Model

State Dependent Models

- ▶ The results above are extreme and probably not realistic: **Two Extensions:**
- ▶ **Proposition 9** : In a Sheshinski-Weiss menu-cost model where frequency of price adjustment is fixed in the very short run, the marginal passthrough from inflation expectations to inflation satisfies: $0 < \phi < 1/2$

General SD Model

State Dependent Models

- ▶ The results above are extreme and probably not realistic: **Two Extensions:**
- ▶ **Proposition 9** : In a Sheshinski-Weiss menu-cost model where frequency of price adjustment is fixed in the very short run, the marginal passthrough from inflation expectations to inflation satisfies: $0 < \phi < 1/2$
- ▶ If the frequency of price adjustment is fixed then inflation on impact satisfies

$$\pi = \bar{h}(\pi_0)(x^*(\pi^e) - \underline{x}(\pi_0))$$

with $\bar{h}$ and $\underline{x}$ held fixed at its previous value.

- ▶ This is the same calculation for the Taylor model except for the value of $x^*(\pi^e) = p^*(\pi^e) - P$. In the Taylor case

$$p^* - P = \frac{1}{2}T\pi^e$$

with $T = T^*(\pi_0)$ fixed at its original value. Now, $T^*(\pi^e)$ is lower.

General SD Model

State Dependent Models

- **Proposition 10** : In a Sheshinski-Weiss menu cost model where in addition to menu costs for changing prices, there are also **fixed costs c_B for changing the pricing bands $(\underline{x}, x^*)$** .

Starting from an steady-state with inflation and expected inflation equal to π_0 , there is an interval of inaction $[\underline{\pi}, \bar{\pi}]$ with $\underline{\pi} < \pi_0 < \bar{\pi}$ such that the the firm maintains its bands unchanged.

Then for any π^e in this interval, inflation remains unchanged, so the passthrough from inflation expectations to inflation is zero.

Conclusions

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- ▶ Step towards in understanding the effect of inflation expectations on pricing and inflation, holding all other determinants of these fixed.
- ▶ Perception that optimizing models imply a 1-to-1 passthrough is generally misleading.
- ▶ The passthrough depends on the pricing model.
- ▶ Much lower values of the passthrough are possible and plausible.
- ▶ Expectations matters to the extent that individual firms “overshoot” their ideal relative price or if the frequency of price increases rises in the short run

Cts Time Limit

Calvo Pricing:

- ▶ This can be done by using the discrete time expression and setting $\beta = e^{-\rho\Delta}$ and $\lambda = e^{-\delta\Delta}$ and then taking the limit as the period length shrink $\Delta \rightarrow 0$ or by setting up the model in continuous time directly. Either way one arrives at

$$\phi = \frac{1}{\rho/\delta + 1},$$

so that passthrough only depends on the ratio ρ/δ .

- ▶ No discounting ($\rho = 0$) or flexible prices ($\delta = \infty$) gives $\phi = 1$.
- ▶ Rigid prices ($\delta = 0$) gives $\phi = 0$.

Cts Time Limit

Taylor Pricing:

- ▶ This can be done by using the discrete time expression and setting $\beta = e^{-\rho\Delta}$ and $N\Delta = 1/\delta$ and then taking the limit as the period length shrink $\Delta \rightarrow 0$ or by setting up the model in continuous time directly. Either way one arrives at

$$\phi(\rho/\delta) = \frac{1}{\rho/\delta} - \frac{1}{e^{\rho/\delta} - 1} \quad (21)$$

so that passthrough only depends on the ratio ρ/δ .

- ▶ No discounting ($\rho = 0$) or flexible prices ($\delta = \infty$) gives $\phi = 1/2$.
- ▶ Rigid prices ($\delta = 0$) gives $\phi = 0$.

General TD Model

General Time Dependent Pricing

- ▶ The price level (in logs) is defined as the average across firms, which equals the weighted average of past reset prices:

$$P_t = \sum_{s=0}^{\infty} \omega_s p_{t-s}^* \quad (22)$$

- ▶ Inflation is then given by

$$\pi_t = P_t - P_{t-1} = \sum_{s=0}^{\infty} \omega_s h_s (p_t^* - p_{t-1-s}^*) \quad (23)$$

inflation is a weighted average of the inflation rate associated with each firm (a fraction $1 - \bar{h}$ firms produce zero inflation).